

MGMT 682: Digital Product Design

Assignment 3: A/B Testing, Continued Development, and Designing Algorithms

Please write your answers as clearly and concisely as possible. Please type your answers. Recall the course policy on AI: you are free to use AI tools subject to disclosure when and how it was used and you are fully responsible for everything you turn in.

You work an online content platform (e.g. YouTube) on the recommendations team. Your team has been tasked with re-designing recommendations that will maximize the user's long-term value/happiness from scratch. You have access to an amazing team of engineers who can train complicated models to predict different variables. That is, let's assume we can perfectly learn conditional expectation functions $E[Y|X = x] = f(x)$. Remember, these are fancy ways to form the best prediction we can about some outcome Y given data x .

1. Designing recommendations. (20 points)
 - a. Recall many recommendation systems predict engagement rates and recommend content in descending order of predicted engagement. Explain (1) why this is a reasonable starting point and (2) whether you think this will be sufficiently aligned with long-term user value. (10 points)
 - b. Your content moderation team worries about users falling down rabbit holes of low-quality content and you have been tasked with improving the recommendation system to prevent this. How might you modify the recommendation algorithm to combat this concern? How do you think this would impact engagement? (5 points)
 - c. How else might you modify your recommendation algorithm to better align with long-term user value? (5 points)
2. We will now test our new algorithm against the existing algorithm using A/B tests. (20 points)
 - a. Describe how you would randomize units into treatment and control. What is the unit of randomization? What share of users would you treat? (5 points)
 - b. How would you test that the randomization worked as expected? Be as specific as you can about what null hypothesis you would test. (5 points)
 - c. How would you evaluate the performance of the new algorithm against the existing algorithm? (5 points)
 - d. Describe a procedure for estimating treatment effect heterogeneity. What subgroups would you focus on? What concerns could arise when you estimate many heterogeneous treatment effects? (5 points)

ANSWER

1. Designing Recommendations

(a) Predicting engagement rates

i. Why predicting engagement rates is a reasonable starting point

Predicting engagement is a reasonable initial strategy because engagement signals (like clicks or time spent) are generally **easy to measure and react quickly to experiments**. This approach effectively captures immediate user interest, which drives short term platform value and aligns with ad revenue models.

ii. Whether this will be sufficiently aligned with long term user value

Relying solely on predicted engagement will **not** be sufficiently aligned with long-term user value or happiness. Engagement is an imperfect, short-term proxy for success. This focus risks promoting sensationalist content, leading to user **regret** and a **narrow content diet**. It can amplify extreme or addictive content, creating **rabbit holes** and reducing **diversity**, ultimately damaging long-term metrics like **retention and satisfaction**.

(b) Modifying the Algorithm to Combat Low Quality Rabbit Holes

i. How might you modify the recommendation algorithm to combat this concern

To prevent low-quality content rabbit holes, the system must become a **quality-aware ranking system**. Using the predictive function, we must predict content quality (Q) in addition to engagement (E). The content rank score (S) should be a weighted combination:

$S = wE * E + wQ * Q$ (Where wE and wQ are the assigned weights for Engagement and Quality, respectively)

Crucially, this modification must involve **structural design changes**, including implementing **diversity constraints**, explicit **exploration strategies**, **safety thresholds**, and rules **limiting sequential similarity in recommendations**.

ii. How would this impact engagement

Implementing a strong weighting toward quality (wQ) will likely result in a **reduction in average immediate engagement**. This is a necessary trade-off to improve long-term user satisfaction.

(c) How else might you modify your recommendation algorithm to better align with long-term user value?

To maximize alignment with long-term user happiness, the platform should redefine its goal using an **Overall Evaluation Criterion (OEC)**.

- i. **Target Predicted Long Term Outcomes:** The algorithm should use the function $E[Y|X=x]=f(x)$ to maximize the **predicted conditional expectation of a long-term metric (Y)**, such as future retention or lifetime value.

- ii. **Incorporate Explicit Satisfaction Signals:** Since behavioral data is insufficient, explicit feedback should be gathered (e.g., surveying users on post worth) and then modeled using the predictive capability to scale **satisfaction signals** across the platform.
- iii. **Establish Penalizing Guardrails:** The OEC must be structured to include **penalties for churn risk**. The OEC score should be reduced based on the predicted probability of a user taking a negative action like unsubscribing due to annoyance.

2. A/B Tests

(a) Randomization unit and share

- i. **Randomization Procedure and Unit:** We must use **random assignment** to divide users into control and treatment groups. The primary unit of randomization should be the **user**, which allows for clear causal identification.
- ii. **Share of Users Treated:** A **50/50 split** is recommended to maximize statistical power.

(b) Testing that the randomization worked as expected

- i. **Testing Procedure:** We conduct **balance tests** on characteristics measured **before** the experiment starts. This requires testing **multiple baseline characteristics**, utilizing **categorical variable tests (like chi-square)**, and **comparing distributions, not only means**, to ensure the groups are comparable.
- ii. **Specific Null Hypothesis:** For any pre treatment characteristic (Y), we test the effect of the treatment indicator (D) using a regression model: $Y = a + b * D + e$. The specific null hypothesis (H_0) we test is that the coefficient on the treatment indicator (b) is zero, indicating no difference in the groups before the intervention: $\{H_0: b = 0\}$

(c) Evaluating the performance of the new algorithm

Evaluation focuses on estimating the **causal impact** (treatment effect, T). Performance is judged based on the predefined **Overall Evaluation Criterion (OEC)**. The treatment effect is calculated as the difference in the average OEC outcome (Y) between the two groups:

$$T = E[Y | D=1] - E[Y | D=0] \text{ (Where } E \text{ represents the average outcome in that group)}$$

(d) Treatment effect heterogeneity

- i. **Procedure for Estimating Heterogeneity:** Heterogeneity is estimated by analyzing how the causal impact varies across user **subgroups**. This is achieved by running separate analyses for defined segments or by including **interaction terms** between the treatment variable and subgroup characteristic in the statistical analysis.
- ii. **Subgroups to Focus On:** Key subgroups should be based on engagement (e.g., light users versus heavy users) and those identified as highly susceptible to consuming **low-quality or sensationalist content**.
- iii. **Concerns with Estimating Many Heterogeneous Treatment Effects:** The primary concern is **Multiple Hypothesis Testing**, which increases the risk of generating **false positives**. Other critical concerns include: **low sample size and variance within subgroups**, **data dredging risk**, **overfitting when exploring many interactions**, and potential **privacy or fairness risks for small groups**.